HOCHSCHULE
RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

Electronic Circuit Design

Slides to the lecture

Chapter 3: Voltage and Current Sources

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Bibliography

[1] Paul Horowitz and Winfield Hill. **The Art of Electronics**. Cambridge University Press, 3rd edition, 2015.

[2] Ulrich Tietze, Christoph Schenk, and Eberhard Gamm. **Electronic Circuits: Handbook for Design and Application**. Springer, 2nd edition, 2008.

[3] Adel S. Sedra and Kenneth C. Smith. **Microelectronic Circuits**. Oxford University Press, 8th edition, 2020.

[4] Walt Jung, editor. **Op Amp Applications Handbook**. Analog Devices / Newnes, 2005.

[5] Op amps for everyone. Technical Report SLOD006B, Texas Instruments, 2002.

[6] Walt Kester, editor. **Data Conversion Handbook**. Analog Devices / Newnes, 2005.

[7] Behzad Razavi. **Principles of Data Conversion System Design**. IEEE Press, 1995.

[8] Robert J. Widlar. New developments in IC voltage regulators. **IEEE Journal of Solid-State Circuits**, 6(1):2–7, 1971.

[9] A. Paul Brokaw. A simple three-terminal IC bandgap reference. **IEEE Journal of Solid-State Circuits**, 9(6):388–393, 1974.

[10] Claude E. Shannon. Communication in the presence of noise. **Proceedings of the IRE**, 37(1):10–21, 1949.

[11] Paul Falstad and Iain Sharp. CircuitJS1 – circuit simulator applet. `https://www.falstad.com/circuit/`, source code `https://github.com/pfalstad/circuitjs1`. accessed October 2026.

[12] Franco Maloberti. **Data Converters**. Springer, 2007.

[13] Robert H. Walden. Analog-to-digital converter survey and analysis. **IEEE Journal on Selected Areas in Communications**, 17(4):539–550, 1999.

[14] IEC. IEC 60751: Industrial platinum resistance thermometers and platinum temperature sensors. International Electrotechnical Commission, 2022.

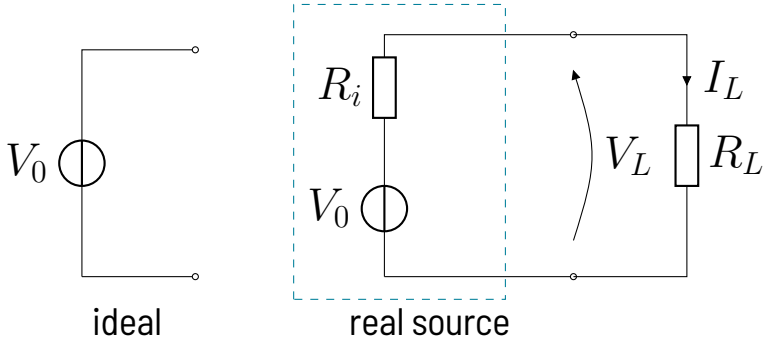
[15] R. P. Sallen and E. L. Key. A practical method of designing RC active filters. **IRE Transactions on Circuit Theory**, 2(1):74–85, 1955.

[16] Charles Kitchin and Lew Counts. A designer's guide to instrumentation amplifiers. Technical report, Analog Devices, 2006.

3 Voltage and Current Sources

3.1 Ideal and Real Voltage Sources

Ideal and real voltage source



- ideal: the voltage is V_0 for every current
- real: the terminal voltage drops with the load current

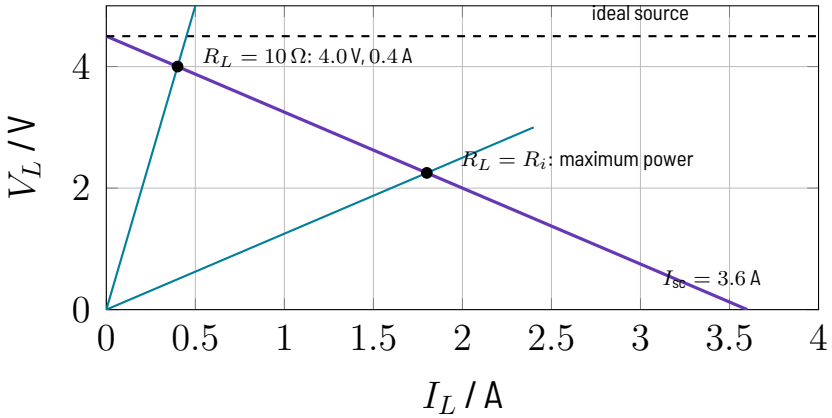
Terminal voltage of a real source

$$V_L = V_0 - R_i\, I_L = V_0\, \frac{R_L}{R_i + R_L}$$

Examples of internal resistances

- AA alkaline cell: $V_0 = 1.5\,\text{V}$, $R_i \approx 0.1\,\Omega \ldots 0.3\,\Omega$ (rises as the cell discharges)
- $9\,\text{V}$ block battery: $R_i \approx 1\,\Omega \ldots 2\,\Omega$
- laboratory power supply in voltage mode: some $\text{m}\Omega$, plus the leads
- voltage divider: $R_i = R_1 \parallel R_2$ (chapter Basics)
- logic output: $20\,\Omega \ldots 50\,\Omega$
- op-amp output with feedback: well below $1\,\Omega$ at low frequencies

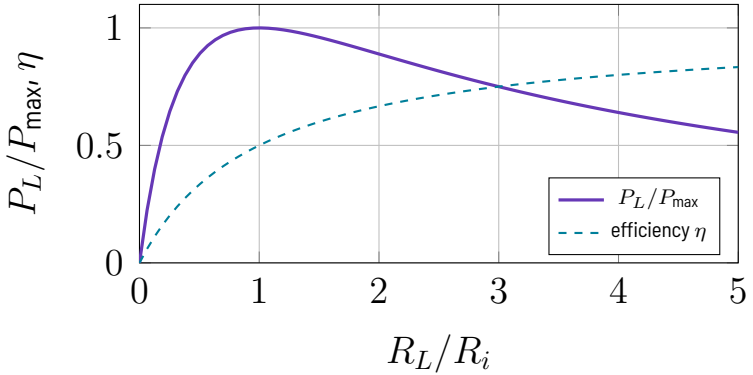
Load line and operating point



- example: $V_0 = 4.5\,\text{V}$, $R_i = 1.25\,\Omega$; load lines of $R_L = 10\,\Omega$ and $R_L = R_i$
- the operating point is the intersection of source line and load line

▶ Falstad: `examples/sources/real_source.txt`

Power matching



- load power $P_L = V_0^2 \, R_L/(R_i + R_L)^2$ is maximum for $R_L = R_i$: $P_{\max} = V_0^2/(4R_i)$
- then half of the power is lost in the source: efficiency $\eta = 50\,\%$
- power supplies and signal sources for measurement: $R_L \gg R_i$ (voltage matching)

Exercise 3.1: Measuring a real voltage source

A battery pack shows 4.5 V without load and 4.0 V with a load of $10\,\Omega$.

a) Calculate the load current and the internal resistance.
b) Calculate the short-circuit current.
c) Which load takes the maximum power from the battery? How large is this power and the efficiency?
d) Check (a) in Falstad and use the slider to find the maximum load power.

Solution 3.1

a) $I_L = 4.0\,\text{V}/10\,\Omega = 0.4\,\text{A}$, $R_i = (4.5\,\text{V} - 4.0\,\text{V})/0.4\,\text{A} = 1.25\,\Omega$.

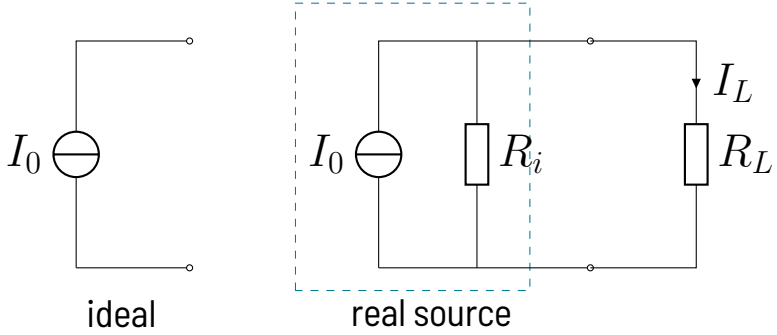
b) $I_{\text{sc}} = V_0/R_i = 4.5\,\text{V}/1.25\,\Omega = 3.6\,\text{A}$.

c) $R_L = R_i = 1.25\,\Omega$: $P_{\text{max}} = (4.5\,\text{V})^2/(4 \cdot 1.25\,\Omega) = 4.05\,\text{W}$, $V_L = 2.25\,\text{V}$, $\eta = 50\,\%$; the battery heats up with the same $4.05\,\text{W}$.

d) ▶ Falstad: `examples/sources/real_source.txt`: `V_ideal` $= 4.50\,\text{V}$, `V_term` $= 4.00\,\text{V}$, $I_L = 0.400\,\text{A}$ (hover over R_L). With the slider at $R_L = 1.25\,\Omega$ the resistor shows $4.05\,\text{W}$.

3.2 Current Sources and Source Transformation

Ideal and real current source

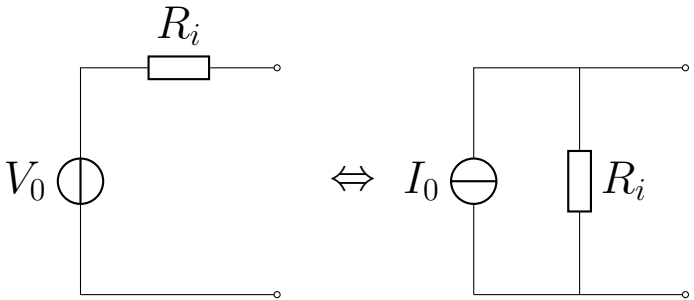


Load current of a real current source

$$I_L = I_0 - \frac{V_L}{R_i} = I_0 \frac{R_i}{R_i + R_L}$$

- a good current source has a **large** internal resistance ($R_i \gg R_L$)
- an ideal current source must never be left open: its voltage would rise without limit
- real current sources work only in a limited voltage range: the **compliance range**

Source transformation



- both have the same open-circuit voltage V_0 and the same short-circuit current $I_0 = V_0/R_i$
- therefore they behave identically for **every** load (Thevenin $\leftrightarrow$ Norton)
- inside, they are different: the power lost in R_i is not the same
- example: $10\,\text{V}$ with $10\,\text{k}\Omega \Leftrightarrow 1\,\text{mA}$ with $10\,\text{k}\Omega$; with $R_L = 4.7\,\text{k}\Omega$ both give $3.20\,\text{V}$

▶ Falstad: `examples/sources/source_transform.txt`

Exercise 3.2: Source transformation

A source consists of $10\,\text{V}$ in series with $10\,\text{k}\Omega$. It feeds $R_L = 4.7\,\text{k}\Omega$.

a) Transform it into a current source with a parallel resistor.
b) Calculate the load voltage with both models.
c) Calculate the power dissipated in the internal resistance for both models. What do you conclude?
d) Check (b) in Falstad.

Solution 3.2

a) $I_0 = 10\,\text{V}/10\,\text{k}\Omega = 1\,\text{mA}$ parallel to $10\,\text{k}\Omega$.

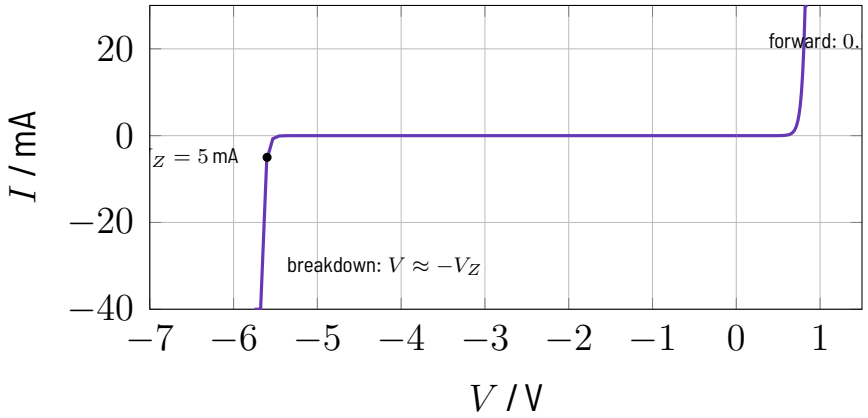
b) Voltage source: $V_L = 10\,\text{V} \cdot 4.7/14.7 = 3.20\,\text{V}$. Current source: $V_L = 1\,\text{mA} \cdot (10\,\text{k}\Omega \parallel 4.7\,\text{k}\Omega) = 1\,\text{mA} \cdot 3.20\,\text{k}\Omega = 3.20\,\text{V}$.

c) Voltage source: $I = 0.680\,\text{mA}$ flows through R_i, $P = 4.63\,\text{mW}$. Current source: R_i sees $3.20\,\text{V}$, $P = 1.02\,\text{mW}$. The models are equivalent only at the terminals, not inside.

d) ▶ Falstad: `examples/sources/source_transform.txt`: `VL_thevenin` = `VL_norton` = $3.20\,\text{V}$.

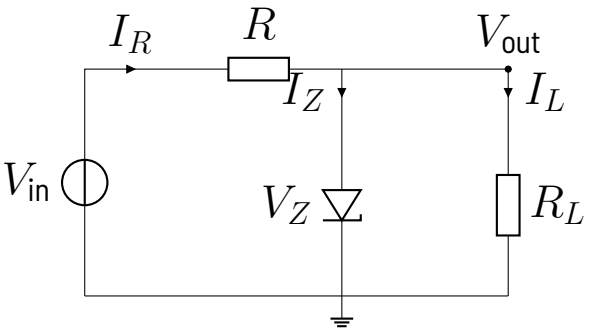
3.3 The Zener Diode Shunt Regulator

Zener diode characteristic



- breakdown voltages 2.4 V … 75 V in E24 steps (BZX55, BZX79: 0.5 W; 1N47xx: 1.3 W)
- datasheet: V_Z at a test current (often 5 mA), dynamic resistance $r_Z = \Delta V / \Delta I$

Shunt regulator



- the resistor takes the voltage difference: $I_R = (V_{\text{in}} - V_Z)/R$
- the Zener takes what the load does not need: $I_Z = I_R - I_L$
- the Zener must keep a minimum current $I_{Z,\text{min}}$ (about 5 mA, or $10\,\%$ of $I_{Z,\text{max}}$)

Designing a shunt regulator

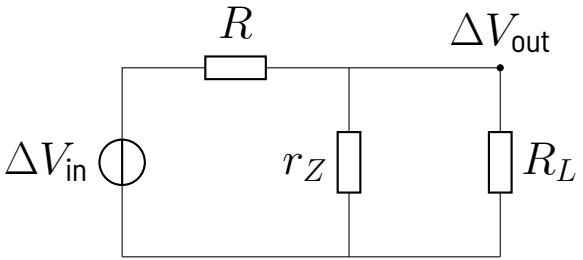
Design card: Zener shunt regulator

1. worst case for regulation: lowest V_{in}, largest load current $\Rightarrow \quad R \leq \dfrac{V_{\text{in,min}} - V_Z}{I_{L,\text{max}} + I_{Z,\text{min}}}$, choose the next **smaller** E-value
2. worst case for the Zener: highest V_{in}, no load $\Rightarrow \quad I_{Z,\text{max}} = \dfrac{V_{\text{in,max}} - V_Z}{R}$, $P_Z = V_Z \, I_{Z,\text{max}}$
3. power of the resistor: $P_R = (V_{\text{in,max}} - V_Z)^2 / R$

- example: $12\,\text{V}$, $5.6\,\text{V}$, $I_L = 0\,\text{mA} \ldots 20\,\text{mA}$: $R \leq 6.4\,\text{V}/25\,\text{mA} = 256\,\Omega \Rightarrow 220\,\Omega$
- the efficiency is poor: the full current flows all the time, even without load

▶ Falstad: `examples/sources/zener_shunt.txt`

Regulation and ripple



- for small changes the Zener is a resistor r_Z: small-signal equivalent circuit
- line regulation: $\dfrac{\Delta V_{\text{out}}}{\Delta V_{\text{in}}} = \dfrac{r_Z \parallel R_L}{R + r_Z \parallel R_L} \approx \dfrac{r_Z}{R}$
- load regulation: output resistance $R_{\text{out}} = R \parallel r_Z \approx r_Z$
- example $r_Z = 20\,\Omega$, $R = 220\,\Omega$: 1 V ripple at the input $\rightarrow 83$ mV at the output

Exercise 3.3: Design of a Zener shunt regulator

A load needs 5.6 V and draws 0 mA … 20 mA. The input voltage is 12 V. Use a 5.6 V Zener diode with $I_{Z,\text{min}} = 5$ mA.

a) Calculate the series resistor and choose an E12 value.
b) Calculate the maximum Zener current and the power of Zener diode and resistor. Which ratings do you choose?
c) Does the circuit still regulate at full load when the input drops to 11 V?
d) Read the output voltage without load and with $R_L = 270\,\Omega$ in Falstad and calculate I_Z.

Solution 3.3

a) $R \leq (12\,\text{V} - 5.6\,\text{V})/(20\,\text{mA} + 5\,\text{mA}) = 256\,\Omega \Rightarrow 220\,\Omega$.

b) Without load: $I_{Z,\text{max}} = 6.4\,\text{V}/220\,\Omega = 29.1\,\text{mA}$, $P_Z = 5.6\,\text{V} \cdot 29.1\,\text{mA} = 163\,\text{mW}$ $\Rightarrow 0.5\,\text{W}$ Zener (BZX79C5V6). $P_R = (6.4\,\text{V})^2/220\,\Omega = 186\,\text{mW} \Rightarrow 0.5\,\text{W}$ resistor.

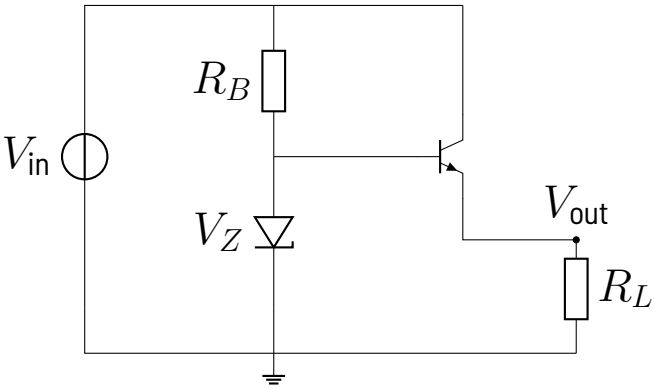
c) $I_R = (11\,\text{V} - 5.6\,\text{V})/220\,\Omega = 24.5\,\text{mA}$, $I_Z = 4.5\,\text{mA}$: just below $I_{Z,\text{min}}$; the voltage drops slightly, but the circuit still works. For a safe margin choose $200\,\Omega$ (E24).

d) ▶ Falstad: `examples/sources/zener_shunt.txt`

`Vout_open` $= 5.65\,\text{V}$ ($I_Z = I_R = 28.9\,\text{mA}$), `Vout_load` $= 5.61\,\text{V}$, $I_L = 5.61\,\text{V}/270\,\Omega = 20.8\,\text{mA}$, $I_R = (12 - 5.61)\,\text{V}/220\,\Omega = 29.0\,\text{mA}$, $I_Z = 8.3\,\text{mA}$.

3.4 Series Regulators

Emitter follower with Zener diode



Emitter follower regulator

$$V_{\text{out}} = V_Z - V_{BE} \approx V_Z - 0.65\,\text{V} \qquad I_B = \frac{I_L}{\beta + 1}$$

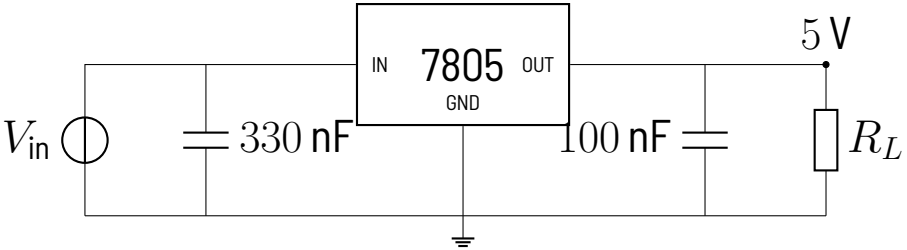
- the Zener sees only I_B: load changes hardly affect V_Z

▶ Falstad: `examples/sources/series_regulator.txt`

Properties of the emitter follower regulator

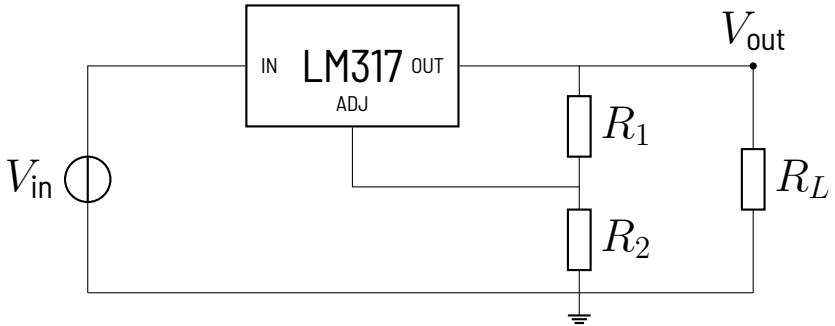
- output resistance $\approx r_e + (R_B \parallel r_Z)/(\beta + 1)$ with $r_e = V_T/I_E$ ($0.25\,\Omega$ at 100 mA): much smaller than with the Zener alone
- V_{BE} depends on current and temperature (-2 mV/K): V_{out} is only accurate to about 0.1 V
- transistor power: $P_Q = (V_{\text{in}} - V_{\text{out}}) \cdot I_L$ (heat sink!)
- design of R_B: $\dfrac{V_{\text{in,min}} - V_Z}{R_B} \geq I_{B,\text{max}} + I_{Z,\text{min}}$
- no short-circuit protection: a short at the output destroys the transistor

Integrated fixed regulator 78xx



- 7805, 7812, ...: fixed output voltage, 1 A (TO-220), 0.1 A (78L05); 79xx for negative voltages
- dropout about 2 V: $V_{\text{in}} \geq V_{\text{out}} + 2$ V; LDO regulators: 0.1 V ... 0.5 V
- internal current limit and thermal shutdown; capacitors close to the pins for stability

Adjustable regulator LM317



LM317

$$V_{\text{out}} = 1.25\,\text{V}\left(1 + \frac{R_2}{R_1}\right) + I_{\text{ADJ}}\,R_2 \qquad I_{\text{ADJ}} \approx 50\,\mu\text{A}$$

- the regulator keeps $1.25\,\text{V}$ between OUT and ADJ; $R_1 = 240\,\Omega$ sets $5.2\,\text{mA}$, enough for the minimum load

Exercise 3.4: Emitter follower regulator

A Zener diode $5.6\,\text{V}$ with $R_B = 1\,\text{k}\Omega$ drives the base of an npn transistor ($\beta = 100$) from $V_{\text{in}} = 12\,\text{V}$. The load is $R_L = 47\,\Omega$.

a) Estimate V_{out} and the load current with $V_{BE} = 0.7\,\text{V}$ (high current).

b) Calculate base current and Zener current.

c) Calculate the power of the transistor. Is a TO-92 transistor ($0.6\,\text{W}$) sufficient?

d) Which series resistor would a Zener shunt regulator need for the same load? Compare the idle current.

e) Compare with Falstad for $R_L = 47\,\Omega$ and $470\,\Omega$.

Solution 3.4

a) $V_{\text{out}} \approx 5.6\,\text{V} - 0.7\,\text{V} = 4.9\,\text{V}$, $I_L = 4.9\,\text{V}/47\,\Omega = 104\,\text{mA}$.

b) $I_B = 104\,\text{mA}/101 = 1.03\,\text{mA}$; $I_{R_B} = 6.4\,\text{V}/1\,\text{k}\Omega = 6.4\,\text{mA}$, $I_Z = 5.4\,\text{mA}$.

c) $P_Q = (12\,\text{V} - 4.9\,\text{V}) \cdot 104\,\text{mA} = 0.74\,\text{W}$: too much for TO-92; use a BD139 (TO-126) with a small heat sink.

d) $R \leq 6.4\,\text{V}/(104\,\text{mA} + 5\,\text{mA}) = 59\,\Omega \Rightarrow 56\,\Omega$. Without load the Zener takes $114\,\text{mA}$ ($0.64\,\text{W}$); the series regulator only draws $6.4\,\text{mA}$.

e) ▶ Falstad: `examples/sources/series_regulator.txt`: $R_L = 47\,\Omega$: $V_{\text{out}} = 4.89\,\text{V}$, $I_L = 104\,\text{mA}$, $V_Z = 5.60\,\text{V}$; $R_L = 470\,\Omega$: $V_{\text{out}} = 4.95\,\text{V}$, $V_Z = 5.61\,\text{V}$.

Exercise 3.5: Adjustable regulator LM317

a) Design an LM317 regulator for $5\,\text{V}$ with $R_1 = 240\,\Omega$. Choose R_2 from E24 and calculate the actual output voltage (with $I_{\text{ADJ}} = 50\,\mu\text{A}$).

b) Which input voltage is needed at least (dropout $2\,\text{V}$)? What is the power of the LM317 at $12\,\text{V}$ input and $300\,\text{mA}$?

c) The LM317 is used as a current source for an LED string with $20\,\text{mA}$. Calculate the resistor.

Solution 3.5

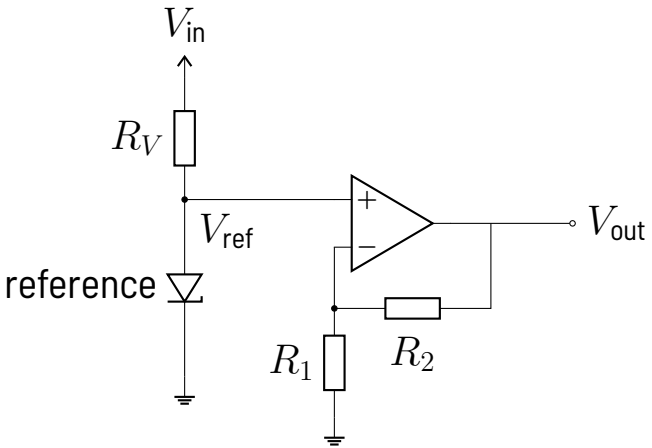
a) $R_2 = R_1\,(5/1.25 - 1) = 3\,R_1 = 720\,\Omega \Rightarrow$ E24: $750\,\Omega$. $V_{\text{out}} = 1.25\,\text{V}\,(1 + 750/240) + 50\,\mu\text{A} \cdot 750\,\Omega = 5.16\,\text{V} + 0.04\,\text{V} = 5.20\,\text{V}$. (With E96: $R_2 = 715\,\Omega$ gives $4.97\,\text{V}$.)

b) $V_{\text{in}} \geq 7.2\,\text{V}$. $P = (12\,\text{V} - 5.2\,\text{V}) \cdot 0.3\,\text{A} = 2.0\,\text{W}$: TO-220 with heat sink.

c) $R = 1.25\,\text{V}/20\,\text{mA} = 62.5\,\Omega \Rightarrow$ E24: $62\,\Omega$ ($20.2\,\text{mA}$); the LM317 then needs $V_{\text{in}} \geq V_{\text{LEDs}} + 1.25\,\text{V} + 2\,\text{V}$.

3.5 Op-Amp Voltage Sources

Reference and non-inverting amplifier



Op-amp voltage source

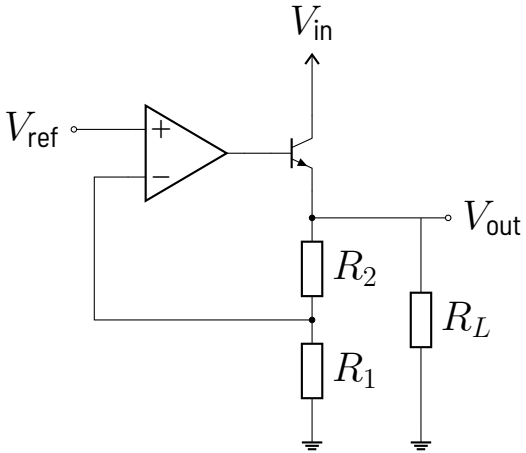
$$V_{\text{out}} = V_{\text{ref}}\left(1 + \frac{R_2}{R_1}\right) \qquad R_{\text{out}} \approx \frac{r_o}{1 + A\beta}, \; \beta = \frac{R_1}{R_1 + R_2}$$

- the reference is not loaded; V_{out} is as accurate as V_{ref} and the ratio R_2/R_1

Output resistance and output current

- example: $r_o = 50\,\Omega$, $A = 10^5$, $R_1 = R_2$: $R_{\text{out}} = 50\,\Omega/(1+0.5\cdot 10^5) = 1\,\text{m}\Omega$
  - 10 mA more load current changes V_{out} by only 10 µV
- but: the op-amp itself delivers only some 10 mA
- and: V_{out} must stay inside the output swing of the op-amp

More current: transistor inside the loop



- npn emitter follower as **booster**: the op-amp delivers only $I_B = I_L/(\beta + 1)$
- V_{BE} lies inside the loop: the op-amp output moves to $V_{\text{out}} + V_{BE}$, V_{out} stays exact
- transistor power $(V_{\text{in}} - V_{\text{out}}) \cdot I_L$; the op-amp output must reach $V_{\text{out}} + V_{BE}$

Example: $5.00\,\text{V}$ from $12\,\text{V}$, up to $100\,\text{mA}$

- reference LM4040-2.5 ($2.500\,\text{V}$, $0.1\,\%$) with $R_V = 4.7\,\text{k}\Omega$: $2.0\,\text{mA}$
- $G = 2$: $R_1 = R_2 = 10\,\text{k}\Omega$ ($0.1\,\%$) $\Rightarrow V_{\text{out}} = 5.00\,\text{V}$
- booster BD139 ($\beta \approx 100$): op-amp output $5.7\,\text{V}$ and $1\,\text{mA}$ at $100\,\text{mA}$ load

	$R_L = 470\,\Omega$	$R_L = 47\,\Omega$
emitter follower with Zener (section 3.4)	$4.95\,\text{V}$	$4.89\,\text{V}$
op-amp and booster	$5.000\,\text{V}$	$5.000\,\text{V}$

▶ Falstad: `examples/sources/opamp_voltage_source.txt`

Exercise 3.6: Op-amp voltage source for $3.3\,\mathrm{V}$

A $3.3\,\mathrm{V}$ supply for a sensor board is built from a $2.5\,\mathrm{V}$ reference, an op-amp and an npn booster ($\beta = 100$, $V_{BE} = 0.75\,\mathrm{V}$). The input voltage is $12\,\mathrm{V}$, the load current up to $200\,\mathrm{mA}$.

a) Choose R_2 for $R_1 = 10\,\mathrm{k\Omega}$ from E24 and from E96. Calculate the output voltages.

b) Which voltage and current must the op-amp output deliver at full load?

c) Calculate the power of the transistor. Compare with the power delivered to the load.

d) In the Falstad example, turn the R_L slider from $1\,\mathrm{k\Omega}$ to $20\,\Omega$. What happens to the output voltage, and what would limit the output current in a real circuit?

Solution 3.6

a) $R_2 = R_1\,(3.3/2.5 - 1) = 3.2\,\text{k}\Omega$. E24: $3.3\,\text{k}\Omega$, $V_\text{out} = 2.5\,\text{V} \cdot 1.33 = 3.325\,\text{V}$ ($0.8\,\%$); E96: $3.24\,\text{k}\Omega$, $V_\text{out} = 3.310\,\text{V}$ ($0.3\,\%$).

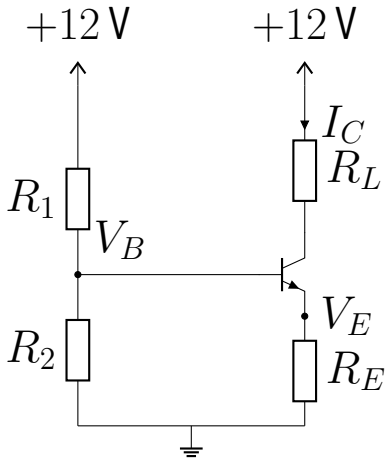
b) $V_\text{out} + V_{BE} \approx 4.06\,\text{V}$; $I_B = 200\,\text{mA}/101 = 1.98\,\text{mA}$ (the divider takes only $0.25\,\text{mA}$ more from the emitter).

c) $P_Q = (12\,\text{V} - 3.31\,\text{V}) \cdot 0.2\,\text{A} = 1.74\,\text{W}$, the load gets only $0.66\,\text{W}$: efficiency $28\,\%$. A heat sink is needed; a switching regulator would be the better choice for this current.

d) ▶ Falstad: `examples/sources/opamp_voltage_source.txt`: `Vout_100mA` stays at $5.000\,\text{V}$ over the whole slider range ($250\,\text{mA}$ at $20\,\Omega$). The simulation has no current limit; in reality the transistor power ($1.75\,\text{W}$), its safe operating area and the op-amp output current (I_L/β) limit the current.

3.6 Transistor Current Sources

BJT current source



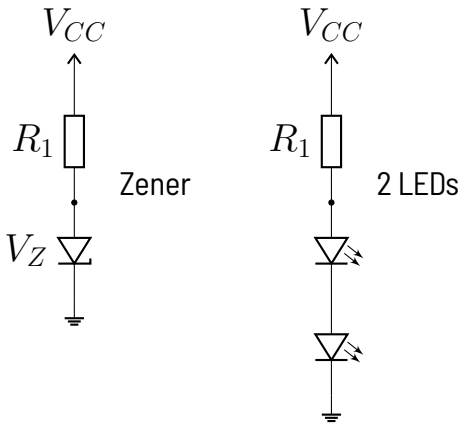
BJT current source

$$I_C \approx I_E = \frac{V_B - V_{BE}}{R_E} \approx \frac{V_B - 0.65\,\text{V}}{R_E}$$

- the emitter follows the base: $V_E = V_B - V_{BE}$ is fixed, so the emitter current is fixed
- the load sits in the collector and may vary within the compliance range

Compliance range and base voltage

- the transistor must stay active: $V_{CE} \geq$ about $0.3\,\mathrm{V}$, i.e. $V_C \geq V_E + 0.3\,\mathrm{V}$
- compliance: $R_{L,\mathrm{max}} = \dfrac{V_{CC} - V_E - 0.3\,\mathrm{V}}{I_C}$; a larger load saturates the transistor and the current drops
- stiff base voltage: divider current $\geq 10\,I_B$; base current error $I_B \cdot (R_1 \parallel R_2)$
- temperature: V_{BE} drops by $2\,\mathrm{mV/K}$; a large V_E ($1\,\mathrm{V}$ and more) makes the current stable

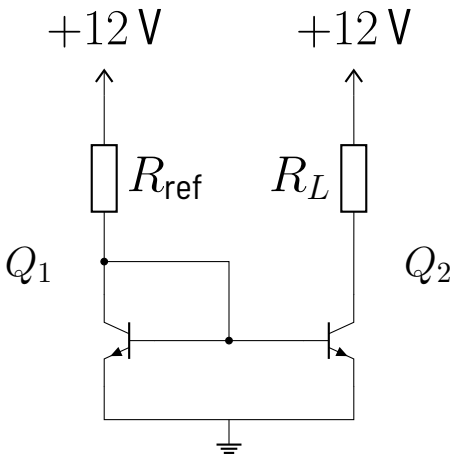


Example: 1 mA current source

- $V_{CC} = 12\,\text{V}$, $R_1 = 10\,\text{k}\Omega$, $R_2 = 2.2\,\text{k}\Omega$: $V_B = 2.16\,\text{V}$ (unloaded)
- $R_E = 1.5\,\text{k}\Omega$: $I_E = (2.16\,\text{V} - 0.65\,\text{V})/1.5\,\text{k}\Omega = 1.0\,\text{mA}$
- loads $1\,\text{k}\Omega$ and $4.7\,\text{k}\Omega$: the same current; $15\,\text{k}\Omega$ is outside the compliance range

▶ Falstad: `examples/sources/bjt_current_source.txt`

Current mirror



- Q_1 is connected as a diode: it sets V_{BE} for the reference current
- Q_2 has the same V_{BE} and therefore the same collector current: $I_{\text{out}} = I_{\text{ref}}\,\beta/(\beta+2)$
- $I_{\text{ref}} = (V_{CC} - V_{BE})/R_{\text{ref}}$; with $11\,\text{k}\Omega$: $1.04\,\text{mA} \rightarrow 1.02\,\text{mA}$

▶ Falstad: `examples/sources/current_mirror.txt`

Exercise 3.7: BJT current source for a resistive sensor

A resistive sensor ($R_S = 1.0\,\text{k}\Omega \ldots 1.4\,\text{k}\Omega$) shall be excited with 1 mA by a BJT current source at $V_{CC} = 12$ V ($V_{BE} = 0.65$ V, $\beta = 100$).

a) The base voltage comes from $R_1 = 10\,\text{k}\Omega$, $R_2 = 2.2\,\text{k}\Omega$. Calculate V_B with and without the base current, and choose R_E from E12.

b) Calculate the voltage across the sensor at $1.0\,\text{k}\Omega$ and $1.4\,\text{k}\Omega$, and the compliance range.

c) The base divider is replaced by two red LEDs in series ($V_F = 1.8$ V each) fed by $4.7\,\text{k}\Omega$. Recalculate R_E.

d) Read the collector currents for $1\,\text{k}\Omega$, $4.7\,\text{k}\Omega$ and $15\,\text{k}\Omega$ in Falstad.

Solution 3.7

a) Unloaded: $V_B = 12\,\text{V} \cdot 2.2/12.2 = 2.16\,\text{V}$. $I_B \approx 10\,\mu\text{A}$ through $R_1 \parallel R_2 = 1.8\,\text{k}\Omega$ lowers it by $18\,\text{mV}$: $V_B = 2.15\,\text{V}$. $R_E = (2.15\,\text{V} - 0.65\,\text{V})/1\,\text{mA} = 1.5\,\text{k}\Omega$ (E12).

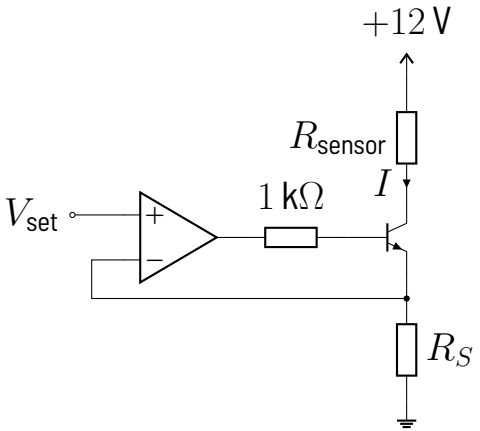
b) $V_S = 1.0\,\text{V}$ and $1.4\,\text{V}$. With $V_E = 1.5\,\text{V}$ and $V_{CE} \geq 0.3\,\text{V}$: $R_{L,\text{max}} = (12 - 1.5 - 0.3)\,\text{V}/1\,\text{mA} = 10.2\,\text{k}\Omega$; the sensor is far inside.

c) $V_B = 3.6\,\text{V}$, $R_E = (3.6\,\text{V} - 0.65\,\text{V})/1\,\text{mA} = 2.95\,\text{k}\Omega \Rightarrow$ E24 $3.0\,\text{k}\Omega$ ($0.98\,\text{mA}$). V_B no longer depends on V_{CC}.

d) ▶ Falstad: `examples/sources/bjt_current_source.txt`: $V_B = 2.15\,\text{V}$, $V_E = 1.55\,\text{V}$, $I_C = 1.023\,\text{mA}$ for $1\,\text{k}\Omega$ and $4.7\,\text{k}\Omega$ ($V_C = 7.19\,\text{V}$), $0.71\,\text{mA}$ for $15\,\text{k}\Omega$ (saturated).

3.7 Op-Amp Current Sources and Sensor Excitation

Op-amp and transistor

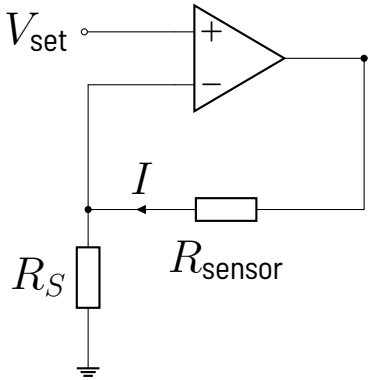


Op-amp current source

$$V_{R_S} = V_{\text{set}} \quad \Rightarrow \quad I_E = \frac{V_{\text{set}}}{R_S} \qquad I = I_C = I_E \frac{\beta}{\beta + 1}$$

- V_{BE} and its drift are inside the loop and drop out; the base current is lost ($1\,\%$ at $\beta = 100$, a MOSFET avoids it)

Floating load in the feedback path

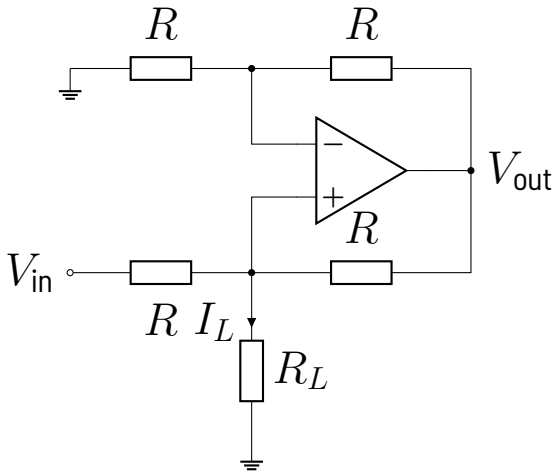


$$V_{\text{out}} = V_{\text{set}}\,(1 + R_{\text{sensor}}/R_S)$$

- non-inverting amplifier: the current V_{set}/R_S through R_S must flow through the sensor
- exact current (no base current); but neither sensor terminal is at ground
- compliance: V_{out} must stay inside the op-amp output range

▶ Falstad: `examples/sources/opamp_current_source.txt`

Howland current pump (grounded load)



- four equal resistors: $I_L = V_{\text{in}}/R$, independent of R_L (within the compliance range)
- the resistors must be matched well: a mismatch of $0.1\,\%$ limits the output resistance

Sensor excitation: why 1 mA?

- resistive temperature sensor (Pt1000 like): $R(T) = R_0\,(1 + \alpha T)$, $R_0 = 1\,\text{k}\Omega$, $\alpha = 3.85 \times 10^{-3}/\text{K}$
  - $0\,°\text{C}$: $1000\,\Omega$, $100\,°\text{C}$: $1385\,\Omega$
- with a constant current I: $V_{\text{sensor}} = I \cdot R(T)$ is linear in T: $3.85\,\text{mV/K}$ at $1\,\text{mA}$
- more current gives more signal, but **self-heating**: $\Delta T = P/G_{\text{th}}$
  - $P = I^2 R = (1\,\text{mA})^2 \cdot 1.385\,\text{k}\Omega = 1.39\,\text{mW}$; with a typical $0.4\,\text{K/mW}$: $0.55\,\text{K}$ error
- the signal offset ($1\,\text{V}$ at $0\,°\text{C}$) is removed by a **bridge**

Example: $1\,\text{mA}$ for a sensor

- $V_{\text{set}} = 1.00\,\text{V}$ (from a reference), $R_S = 1.00\,\text{k}\Omega$ ($0.1\,\%$): $I = 1.000\,\text{mA}$
- op-amp and transistor: sensor current $0.990\,\text{mA}$ (base current missing)
- floating sensor in the feedback path: $1.000\,\text{mA}$, $V_{\text{out}} = 1\,\text{V}\,(1 + 1.385) = 2.385\,\text{V}$ at $100\,^\circ\text{C}$
- accuracy of the current = accuracy of V_{set} and R_S

▶ Falstad: `examples/sources/opamp_current_source.txt`